6.7 Electrical Characteristics: OPAx227

at T_A = +25°C, V_S = ±5 V to ±15 V, V_{CM} = V_{OUT} = V_S / 2, and R_L = 10 kΩ connected to V_S / 2 (unless otherwise noted)

PARAMETER		TEST CONDITIONS		MIN	TYP	MAX	UNIT
OFFSET VOLTAGE							
V_{OS}	Input offset voltage	U, P grades			±5	±75	µV
			T_A = –40°C to +85°C			±100	
		UA, PA grades			±10	±200	
			T_A = –40°C to +85°C			±200	
dV_{OS}/dT	Input offset voltage drift	U, P grades	T_A = –40°C to +85°C		±0.1	±0.6	µV/°C
		UA, PA grades	T_A = –40°C to +85°C		±0.3	±2	
PSRR	Power-supply rejection ratio	V_S = ±2.5 V to ±18 V			±0.5	±2	µV/V
			T_A = –40°C to +85°C			±2	
	Long-term drift				0.2		µV/mo
	Channel separation (dual, quad)	DC			0.2		µV/V
		f = 1 kHz, R_L = 5 kΩ			110		dB
INPUT BIAS CURRENT							
I_B	Input bias current				±2.5	±10	nA
		T_A = –40°C to +85°C				±10	
I_{OS}	Input offset current				±2.5	±10	nA
		T_A = –40°C to +85°C				±10	
NOISE							
	Input voltage noise	f = 0.1 Hz to 10 Hz			90		nV_{PP}
					15		nV_{RMS}
e_n	Input voltage noise density	f = 10 Hz			3.5		nV/√Hz
		f = 100 Hz			3		
		f = 1 kHz			3		
i_n	Input current noise density	f = 1 kHz			0.4		pA/√Hz
INPUT VOLTAGE							
V_{CM}	Common-mode voltage			(V–) + 2		(V+) – 2	V
CMRR	Common-mode rejection ratio	(V–) + 2 V < V_{CM} < (V+) – 2 V		120	138		dB
			T_A = –40°C to +85°C	120			
INPUT IMPEDANCE							
	Differential				10 || 12		MΩ || pF
	Common-mode	V_{CM} = (V–) + 2 V to (V+) – 2 V			1 || 3		GΩ || pF

Copyright © 2023 Texas Instruments Incorporated